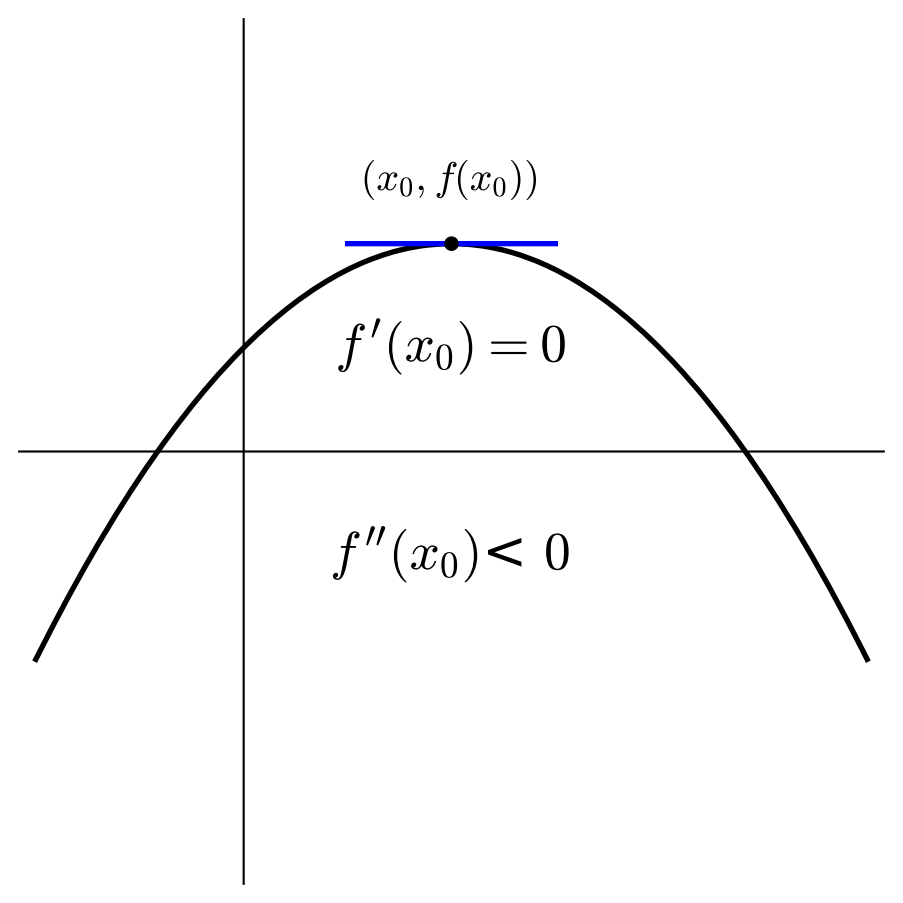




A graph of a function $f(x)$ is shown on a Cartesian coordinate system. The function is represented by a thick black curve that is concave down, forming an inverted parabola. A horizontal blue line segment is drawn tangent to the peak of the curve at a point marked with a black dot. The peak is labeled with the coordinates $(x_0, f(x_0))$. Below the peak, the first derivative is given as $f'(x_0) = 0$, and further down, the second derivative is given as $f''(x_0) < 0$. The horizontal axis is a thin black line, and the vertical axis is also a thin black line.

$$(x_0, f(x_0))$$

$$f'(x_0) = 0$$

$$f''(x_0) < 0$$